

Practical - 5

Practical Title: Registration Form with Frontend Validation and User-Friendly Error Handling

Create a student registration form with HTML5 input types and JavaScript validation for name, email, mobile number, password, confirm password, course, year, gender, and terms acceptance. Use Regular Expression for validations.

A. Theory

1. Are correct input types and attributes used?

Yes. Appropriate HTML5 input types and attributes are used for each field. For example, `email` is used for email, `tel` for mobile number, and `password` for passwords. Attributes such as `required`, `placeholder`, `minlength`, and `maxlength` help restrict invalid input and improve form validation.

2. Are validation errors displayed near the relevant fields?

Yes. Validation errors are displayed near the corresponding input field. This makes it easy for the user to identify and correct the invalid information instead of searching for the error elsewhere on the page.

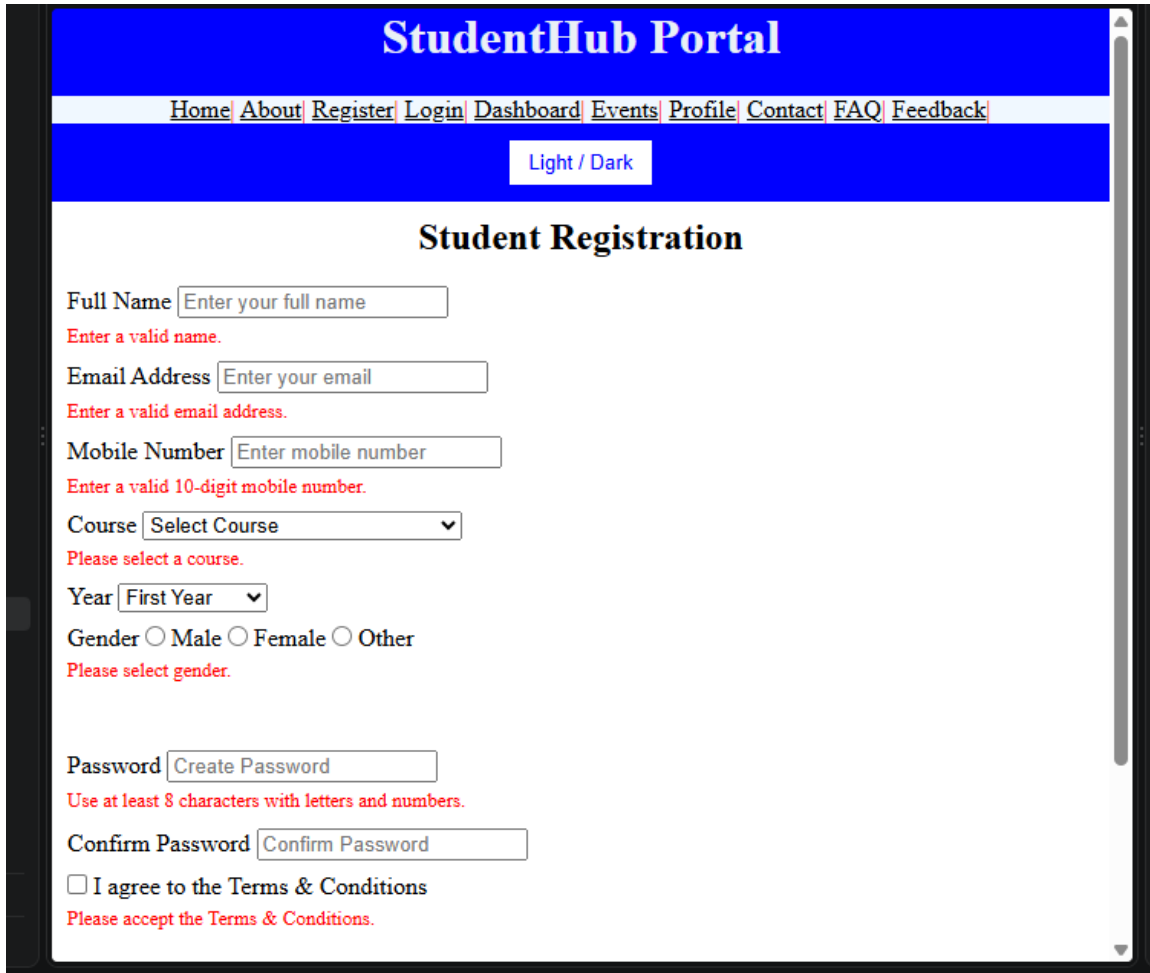
3. Is password strength checked?

Yes. Password strength is checked using JavaScript validation. The password can be evaluated based on conditions such as minimum length, uppercase/lowercase letters, numbers, and special characters. A password-strength indicator can show whether the password is **Weak, Medium, or Strong**.

4. Is the form accessible using keyboard and screen-reader-friendly labels?

Yes. Each form field should have a properly associated `<label>` using the `for` attribute. This allows screen readers to identify the purpose of each input. The form can also be operated using the keyboard by moving between fields with the **Tab** key and activating controls with the keyboard.

B. Implementation/ Output/Result



The screenshot displays the 'StudentHub Portal' with a navigation menu including Home, About, Register, Login, Dashboard, Events, Profile, Contact, FAQ, and Feedback. A 'Light / Dark' toggle is present. The 'Student Registration' form includes fields for Full Name, Email Address, Mobile Number, Course (dropdown), Year (dropdown), Gender (radio buttons), Password, and Confirm Password. Each field has a placeholder and a red error message below it. A checkbox for 'I agree to the Terms & Conditions' is at the bottom.

StudentHub Portal

[Home](#) [About](#) [Register](#) [Login](#) [Dashboard](#) [Events](#) [Profile](#) [Contact](#) [FAQ](#) [Feedback](#)

Light / Dark

Student Registration

Full Name
Enter a valid name.

Email Address
Enter a valid email address.

Mobile Number
Enter a valid 10-digit mobile number.

Course Please select a course.

Year Please select year.

Gender ☐ Male ☐ Female ☐ Other
Please select gender.

Password
Use at least 8 characters with letters and numbers.

Confirm Password

☐ I agree to the Terms & Conditions
Please accept the Terms & Conditions.

The screenshot shows a registration form with the following elements:

- Password field:** A text input with the placeholder "Create Password". Below it, a red error message reads: "Use at least 8 characters with letters and numbers."
- Confirm Password field:** A text input with the placeholder "Confirm Password".
- Terms & Conditions:** A checkbox labeled "I agree to the Terms & Conditions". Below it, a red error message reads: "Please accept the Terms & Conditions."
- Register button:** A button labeled "Register".
- Error summary:** A red error message below the button reads: "Please correct the errors shown above."
- Footer:** A blue bar at the bottom contains the text: "© 2026 StudentHub Portal | Web Development Frameworks".

Description:

The registration form uses HTML5 and JavaScript to provide proper input validation, user-friendly error messages, password-strength checking, and accessible form controls. These features improve the accuracy, security, usability, and accessibility of the StudentHub registration form.

C. GitHub Repository Link:

<https://github.com/vatsal-1950/StudentHUB.git>

D. Challenges Faced:

- **Form validation:** Implementing validation for different fields such as name, email, mobile number, password, and confirm password was challenging.
- **Regular expressions:** Creating and applying suitable regular expressions for email, mobile number, and password validation required careful testing.
- **Error messages:** Displaying clear and user-friendly error messages near the corresponding fields was challenging.
- **Password strength:** Implementing the password-strength indicator and checking multiple password conditions required additional JavaScript logic.
- **Accessibility:** Ensuring that all fields had proper labels and could be accessed using the keyboard required attention to the HTML structure.

- **Debugging:** Testing valid and invalid inputs and correcting JavaScript errors took time during implementation.

E. Learning Outcomes:

- **Create a student registration form** using HTML5 form elements.
- **Apply JavaScript validation** to validate user inputs such as name, email, mobile number, password, and confirm password.
- **Use regular expressions** for validating different types of user input.
- **Display user-friendly error messages** near the relevant form fields.
- **Implement password-strength checking** to improve password security.
- **Design accessible forms** using proper labels and keyboard-friendly controls.
- **Test and debug the form** using both valid and invalid inputs.

Faculty Remarks: (Keep this blank for faculty)